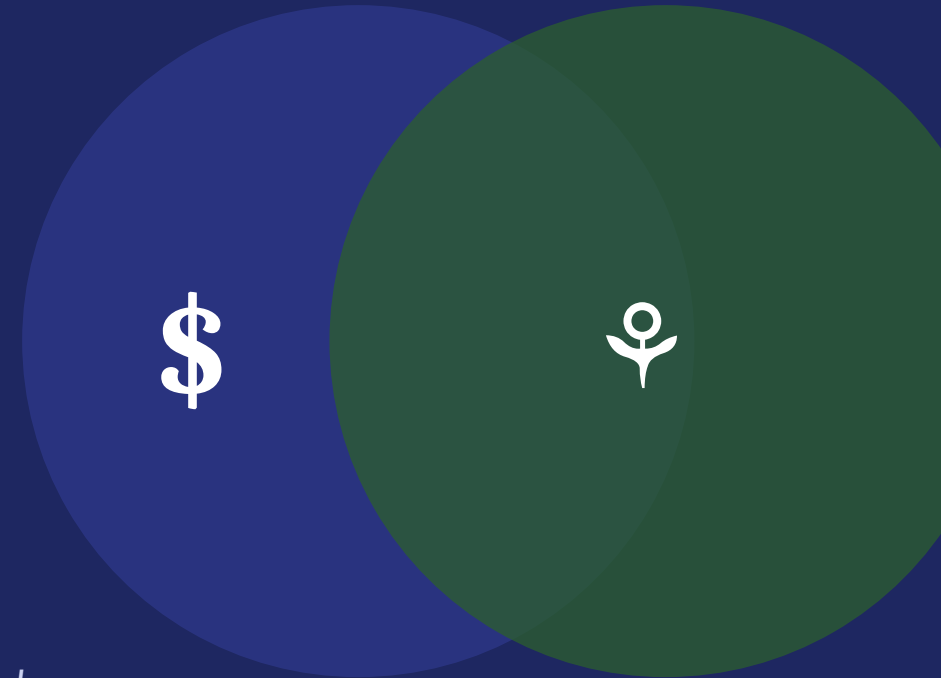


DOUBLE MATERIALITY ASSESSMENT

An ESRS-aligned framework for scoring, ranking, and defending ESG priorities — built and demonstrated on Tata Steel Limited

EXCEL-BASED ANALYTICAL TOOL • FORMULA-DRIVEN • GRI / ESRS ALIGNED

Ishit Yadav • github.com/ishityadav



ESG topics are everywhere. Materiality is not.

CSRD/ESRS now requires companies to justify — not just assert — which ESG topics matter, and from which direction they matter.

1

No shared scoring logic

Most reports rate topics “High / Medium / Low” with no visible method behind the label.

2

Scores aren’t traceable

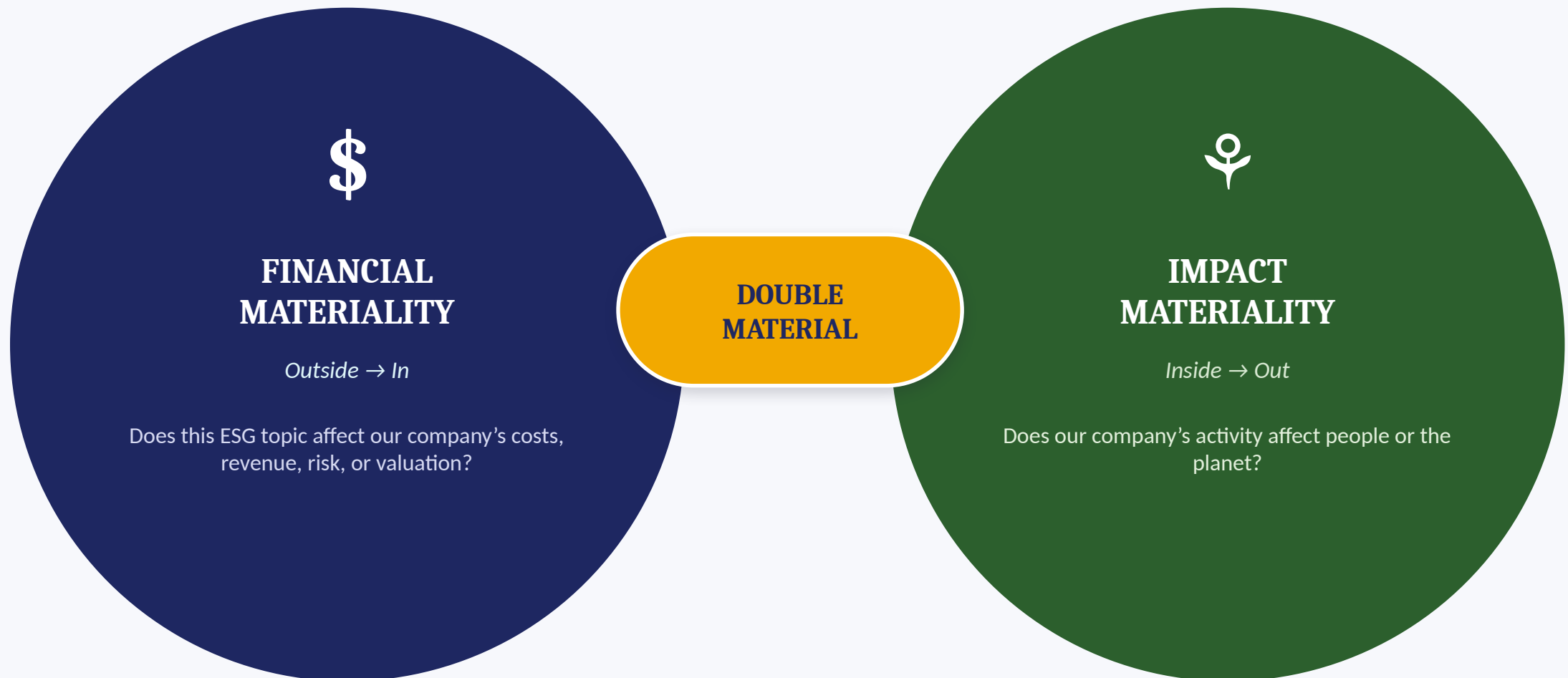
A rating that can’t be traced back to a reason can’t survive an auditor’s or investor’s question.

3

Static outputs

A PDF matrix can’t be recalculated when an assumption, a regulation, or a stakeholder view changes.

Every topic is checked from two directions



A topic must be disclosed under ESRS if it is material under EITHER lens — or both.

Two formulas turn judgment into a number

FINANCIAL MATERIALITY SCORE

Likelihood × Magnitude

max score: 25

IMPACT MATERIALITY SCORE

AVG(Scale, Scope, Irrem.) × Likelihood

max score: 25

Five variables, each scored 1-5, feed the two formulas above

BOTH

Likelihood

How probable is the risk or impact?

FINANCIAL

Magnitude

How big is the financial hit if it happens?

IMPACT

Scale

How severe is the harm to people/planet?

IMPACT

Scope

How widespread is the harm?

IMPACT

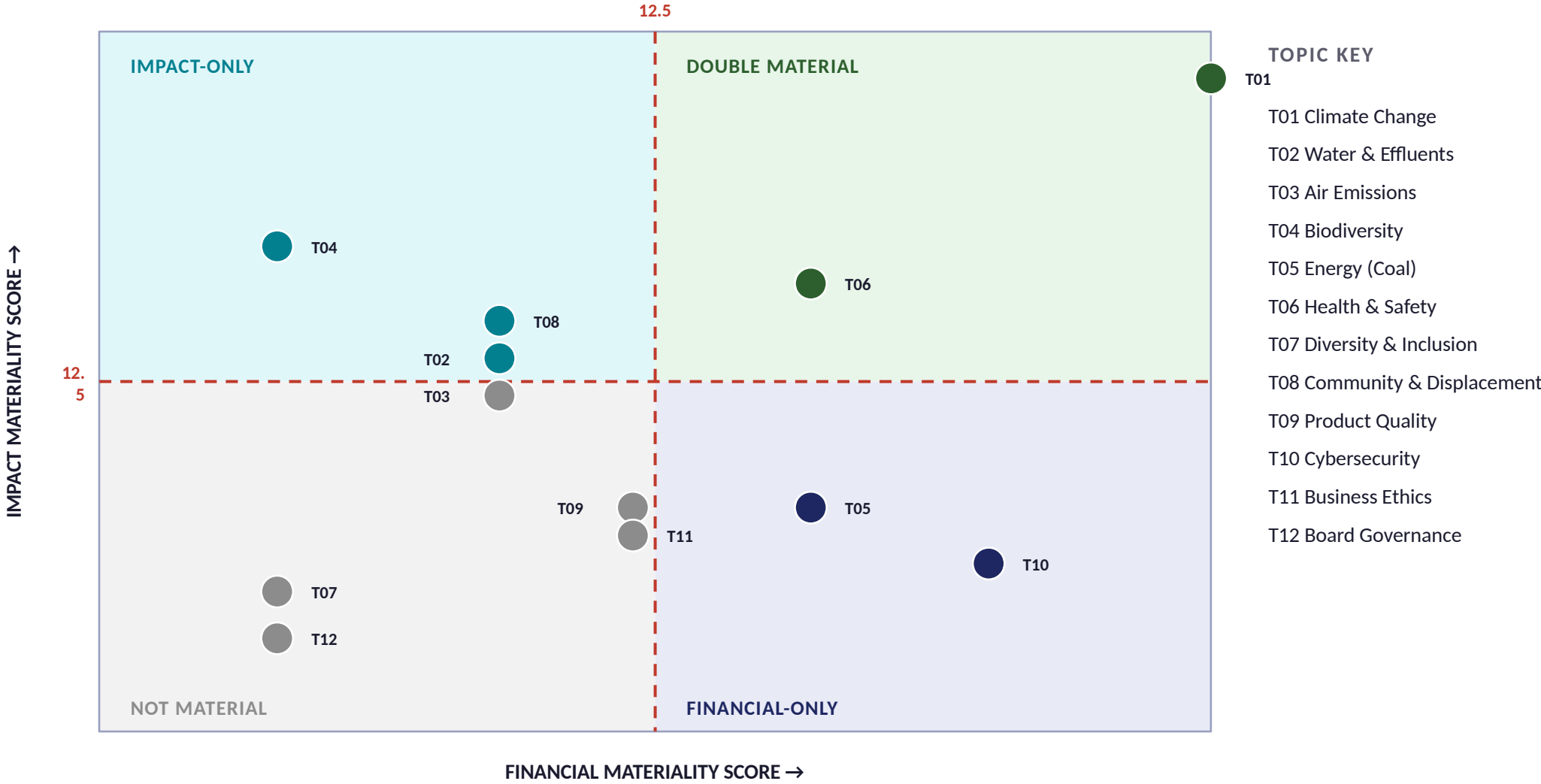
Irremediability

How hard is it to undo the damage?

THE MATRIX

Where Tata Steel’s 12 topics land

Threshold = 12.5 on each axis • scores run 0–25



12 GRI-aligned ESG topics, scored end to end

ENVIRONMENTAL

-  T01
Climate Change (GHG Emissions)
-  T02
Water Consumption & Effluents
-  T03
Air Emissions (Dust, SOx, NOx)
-  T04
Biodiversity & Mine Rehabilitation
-  T05
Energy Management (Coal)

SOCIAL

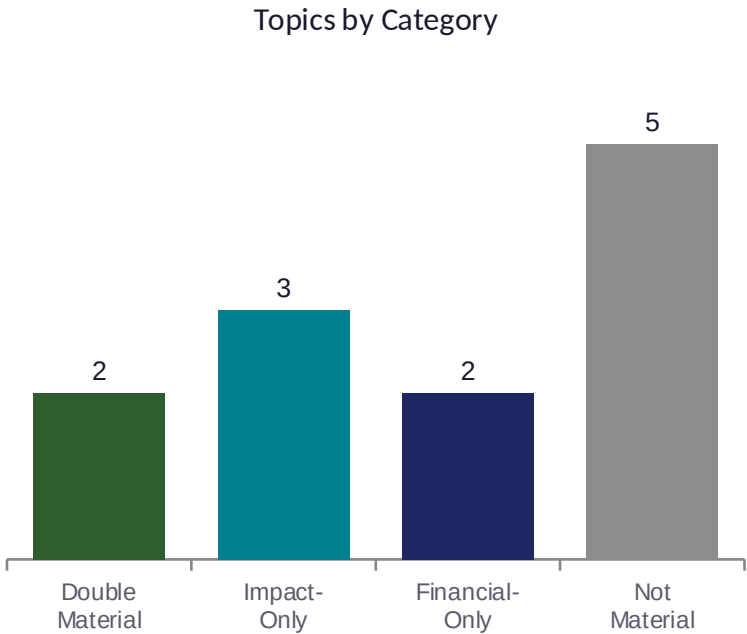
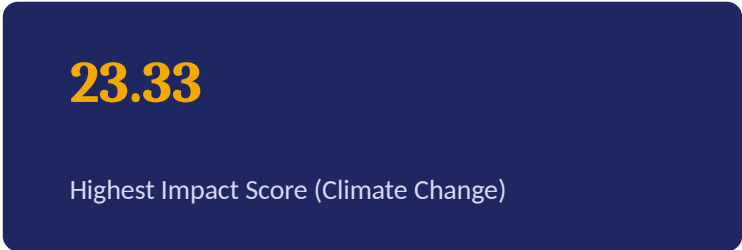
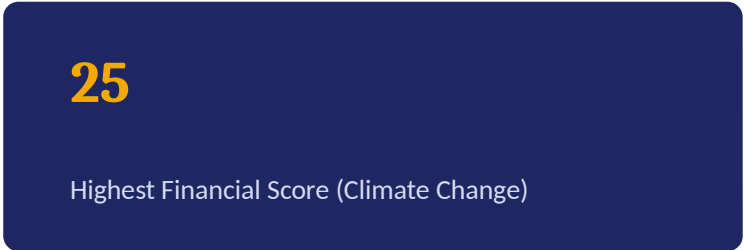
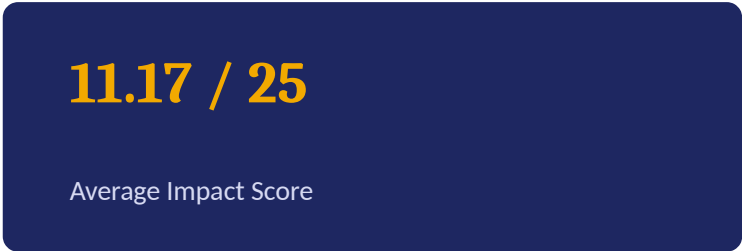
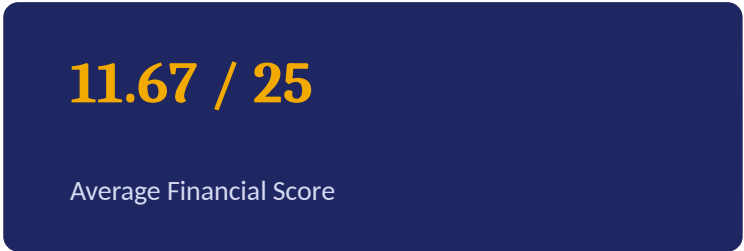
-  T06
Occupational Health & Safety
-  T07
Employee Diversity & Inclusion
-  T08
Community Development & Displacement
-  T09
Product Quality & Customer Safety

GOVERNANCE

-  T10
Cybersecurity Resilience
-  T11
Business Ethics & Anti-Corruption
-  T12
Board Governance & Climate Oversight

RESULTS

The numbers behind the matrix



KEY INSIGHT

A materiality outcome lives or dies at the threshold

Three topics — Air Emissions, Product Quality, and Business Ethics — each scored exactly 12 on one axis. Moving the threshold from 12 to 12.5 flipped all three from “material” to “Not Material.”

THRESHOLD = 12

10 topics classified as material

(2 Double + 4 Impact-Only + 4 Financial-Only)



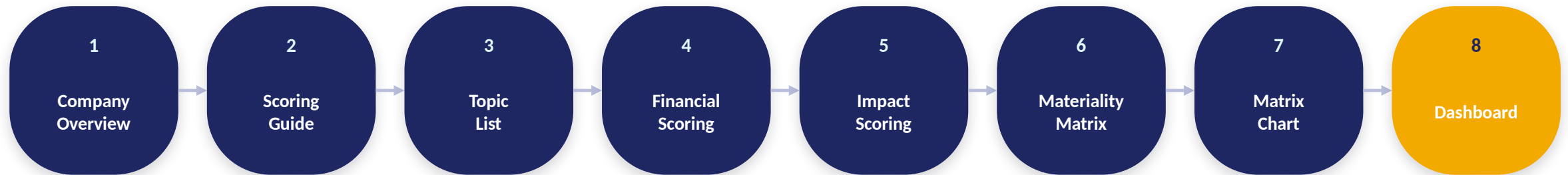
THRESHOLD = 12.5

Only 7 topics remain material — 3 fewer sit inside the boundary

Air Emissions, Product Quality, and Business Ethics move to “Not Material”

The lesson: a materiality threshold is a governance decision, not a technicality — this workbook makes it visible and adjustable in one place, not buried in a PDF.

Eight linked sheets, one live calculation chain



Every downstream sheet uses INDEX/MATCH to pull from the sheet before it — change one input score, and the matrix, chart, and dashboard recalculate automatically.

163

live formulas, zero calculation errors

12.5

adjustable materiality threshold

100%

standard Excel functions — no add-ins, no macros

Built to be trusted, not just to look finished

1

Fully formula-driven

Nothing is a hardcoded output — every score, average, and category label recalculates live from its inputs.

2

Every score has a reason

Each of the 24
Likelihood/Magnitude/Scale/Scope/Irremediability inputs
carries a one-line, auditable justification.

3

Built to be edited

Change one input on the Financial or Impact sheet and the matrix, chart, and dashboard update themselves — no rebuilding.

4

Beginner-friendly by design

A built-in Scoring Guide sheet explains every 1–5 rating in plain English, colour-coded green to red.

From training exercise to reusable template

01

Expand the topic universe

Grow from 12 teaching topics to the full ESRG/GRI sector-standard topic set.

02

Plug in real data

Replace expert-judgment scores with stakeholder surveys, incident logs, and financial exposure data.

03

Sector-agnostic template

Generalize the workbook so any company can swap in its own topics and scores.

04

Companion dashboard

A lightweight Power BI / Python layer for multi-year tracking and peer benchmarking.

Turning ESG judgment into an auditable system.

Full workbook, methodology notes, and documentation available on GitHub.

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